1. Write a Java program for error detecting code using CRC-CCITT (16- bits).

```
import java.io.*;
class Crc
{
	public static void main(String args[]) throws IOException
	{
		BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
		int[ ] data;
		int[  ]div; int[ ]divisor; int[ ]rem;
		int[ ] crc;
		int data_bits, divisor_bits, tot_length;
		System.out.println("Enter number of data bits : ");
		data_bits=Integer.parseInt(br.readLine());
		data=new int[data_bits];
		System.out.println("Enter data bits : ");
		for(int i=0; i<data_bits; i++)
			data[i]=Integer.parseInt(br.readLine());
		System.out.println("Enter number of bits in divisor : ");
		divisor_bits=Integer.parseInt(br.readLine());
		divisor=new int[divisor_bits];
		System.out.println("Enter Divisor bits : ");
		for(int i=0; i<divisor_bits; i++)
			divisor[i]=Integer.parseInt(br.readLine());
		tot_length=data_bits+divisor_bits-1; div=new int[tot_length];
		rem=new int[tot_length];
		crc=new int[tot_length];
		/*------------------ CRC GENERATION-----------------------*/
		for(int i=0;i<data.length;i++) div[i]=data[i];
			System.out.print("Dividend (after appending 0's) are : ");
		for(int i=0; i< div.length; i++)
			System.out.print(div[i]);
		System.out.println();
		for(int j=0; j<div.length; j++)
		{
			rem[j] = div[j];
		}
		rem=divide(div, divisor, rem);
		for(int i=0;i<div.length;i++)   //append dividend and ramainder
		{
			crc[i]=(div[i]^rem[i]);
		}
		System.out.println(); System.out.println("CRC code : ");
		for(int i=0;i<crc.length;i++)
```

```java
			System.out.print(crc[i]);
			/*-------------------ERROR DETECTION---------------------*/
			System.out.println();
			System.out.println("Enter CRC code of "+tot_length+" bits : ");
			for(int i=0; i<crc.length; i++)
				crc[i]=Integer.parseInt(br.readLine());
			for(int j=0; j<crc.length; j++)
			{
				rem[j] = crc[j];
			}
			rem=divide(crc, divisor, rem);
			for(int i=0; i< rem.length; i++)
			{
				if(rem[i]!=0)
				{
					System.out.println("Error"); break;
				}
				if(i==rem.length-1) System.out.println("No Error");
			}
			System.out.println("THANK YOU.... :)");
		}
		static int[] divide(int div[],int divisor[], int rem[])
		{
			int cur=0; while(true)
			{
				for(int i=0;i<divisor.length;i++) rem[cur+i]=(rem[cur+i]^divisor[i]);
				while(rem[cur]==0 && cur!=rem.length-1) cur++;
				if((rem.length-cur)<divisor.length) break;
			}
			return rem;
		}
	}
```